

Lucas Narloch Zabla

Software Engineer

+55 (41) 98719-8655 | lucasnarloch123@gmail.com | Curitiba - PR, Brazil

[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Software Engineer with production experience building backend applications, RESTful APIs, internal tools, and automation solutions in AWS environments. Strong focus on clean and maintainable code, automated testing, logging, troubleshooting, infrastructure automation, and continuous improvement.

TECHNICAL SKILLS

- **Backend:** Node.js, TypeScript, Python, NestJS, FastAPI
- **Databases:** PostgreSQL, MySQL, DynamoDB
- **Cloud & DevOps:** AWS Lambda, S3, SQS, EC2, RDS, IoT Core, CloudWatch, Terraform, GitHub Actions, Docker
- **Frontend:** React, Next.js, TailwindCSS

EXPERIENCE

Backend Developer - SmartLy Fabricação de Dispositivos Inteligentes Ltda.

10/2026 - Present

- Lead the migration from multiple AWS Lambda functions to a centralized Python and FastAPI API, reorganizing the backend into services and repositories and reducing deployment time by 30% and latency by up to 70% on specific endpoints.
- Redesigned AWS network architecture, replacing multiple VPC endpoints with a NAT Gateway and implementing an EC2 bastion for database access, reducing monthly network costs by approximately 64% (US\$90 to US\$32) and expanding Lambda connectivity.
- Lead the restructuring of the MySQL schema for a production IoT platform, remodeling entities, relationships, and migration flows to better reflect the domain, improve data consistency, and preserve integrity and compatibility during transition.

Backend Developer Intern - SmartLy Fabricação de Dispositivos Inteligentes Ltda.

09/2024 - 09/2026

- Implemented asynchronous processing with Amazon SQS for event-driven communication between services, processing peaks above 40,000 notifications per day and removing blocking operations from the main request flow.
- Developed an endpoint for data migration between IoT devices, integrating MySQL, DynamoDB, S3, and AWS IoT Core through MQTT and supporting the migration of thousands of historical records per operation.
- Optimized Python AWS Lambda functions, reducing the average number of errors every 5 minutes from 113 to 5 (95.6%) and significantly increasing production system reliability.
- Built an automated test suite for FastAPI APIs with pytest and Testcontainers, including 120 integration tests and 300 unit tests, improving delivery reliability and reducing production regression risk.

PROJECTS

Sunlog.dev

Production web platform designed for developers to document projects, technical decisions, and development progress. Built with a RESTful backend, PostgreSQL, automated infrastructure, authentication, background processing, and production observability.

The platform includes 33 HTTP endpoints, more than 100 automated tests, JWT-based authentication, asynchronous processing, CI/CD pipelines with GitHub Actions, and monitoring with Prometheus and Grafana.

Technologies: NestJS, TypeScript, PostgreSQL, JWT, Docker, Nginx, AWS, GitHub Actions, Prometheus, Grafana, Next.js

[Code](#) | [Live demo](#)

PocScript

Programming language and compiler implemented in C, covering the complete compilation pipeline, including lexical analysis, AST parsing, semantic analysis, custom intermediate representation generation, LLVM IR emission, and native executable generation.

Includes a custom runtime based on Linux system calls, a static type system, scope resolution, arrays, pointers, error handling, and 196 unit and integration tests.

Technologies: C, LLVM IR, NASM, Make, Linux system calls, Compiler Design

[Code](#)

EDUCATION

LANGUAGES

- **Portuguese:** Native
- **English:** Professional Working Proficiency

Bachelor of Computer Science - PUCPR

02/2025 - 12/2028

Full scholarship recipient. Coursework focused on algorithms, data structures, object-oriented programming, operating systems, computer networks, databases, and software engineering.